EXP1 : SIMULATION AND COMPARATIVE ANALYSIS OF FDMA, TDMA AND CDMA: RESOURCE ALLOCATION, USER CAPACITY AND MOBILE - GENERATION CONTEXT

MATLAB CODE:

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clc;
clear;
close all;
%% FDMA, TDMA AND CDMA COMPARATIVE SIMULATION
% Common system parameters
B = 1.25e6;             % Total bandwidth (Hz)
Rb = 9.6e3;             % Data rate per user (bps)
SNR_dB = 7;             % Target Eb/N0 in dB
SNR = 10^(SNR_dB/10);
%% 1. FDMA SIMULATION
% Each user gets a separate frequency channel
channelBW = 30e3;       % Channel bandwidth (Hz)
FDMA_users = floor(B/channelBW);
fprintf('\n------ FDMA RESULTS ------\n');
fprintf('Total Bandwidth: %.2f MHz\n',B/1e6);
fprintf('Channel Bandwidth: %.2f kHz\n',channelBW/1e3);
fprintf('Maximum Users: %d\n',FDMA_users);
%% 2. TDMA SIMULATION
% Users share a carrier in different time slots
carrierBW = 200e3;      % Carrier bandwidth (Hz)
slots = 8;              % Number of time slots per carrier
carriers = floor(B/carrierBW);
TDMA_users = carriers * slots;
fprintf('\n------ TDMA RESULTS ------\n');
fprintf('Carrier Bandwidth: %.2f kHz\n',carrierBW/1e3);
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fprintf('Time Slots: %d\n',slots);
fprintf('Available Carriers: %d\n',carriers);
fprintf('Maximum Users: %d\n',TDMA_users);
%% 3. CDMA SIMULATION
% All users share the same frequency and time
% Users are separated using spreading codes
processingGain = B/Rb;
CDMA_users = floor(1 + processingGain/SNR);
fprintf('\n------ CDMA RESULTS ------\n');
fprintf('Processing Gain: %.2f\n',processingGain);
fprintf('Target Eb/N0: %.2f dB\n',SNR_dB);
fprintf('Estimated Maximum Users: %d\n',CDMA_users);
%% 4. CAPACITY COMPARISON
capacity = [FDMA_users, TDMA_users, CDMA_users];
techniques = {'FDMA','TDMA','CDMA'};
figure('Name','User Capacity Comparison');
bar(capacity);
set(gca,'XTickLabel',techniques);
ylabel('Number of Users');
xlabel('Multiple Access Techniques');
title('User Capacity Comparison');
grid on;
%% 5. RESOURCE ALLOCATION COMPARISON
figure('Name','Resource Allocation');
% FDMA frequency allocation
subplot(3,1,1);
imagesc(ones(FDMA_users,10));
title('FDMA - Separate Frequency Channels');
ylabel('Users');
xlabel('Time Slots');
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colorbar;
% TDMA time allocation
subplot(3,1,2);
imagesc(repmat(1:slots,carriers,1));
title('TDMA - Different Time Slots');
ylabel('Carriers');
xlabel('Time Slots');
colorbar;
% CDMA code allocation
subplot(3,1,3);
imagesc(ones(CDMA_users,10));
title('CDMA - Shared Frequency with Different Codes');
ylabel('Users');
xlabel('Time Slots');
colorbar;
%% 6. SPECTRAL EFFICIENCY
FDMA_eff = (FDMA_users*Rb)/B;
TDMA_eff = (TDMA_users*Rb)/B;
CDMA_eff = (CDMA_users*Rb)/B;
efficiency = [FDMA_eff TDMA_eff CDMA_eff];
figure('Name','Spectral Efficiency');
bar(efficiency);
set(gca,'XTickLabel',techniques);
ylabel('Spectral Efficiency (bps/Hz)');
title('Spectral Efficiency Comparison');
grid on;
%% 7. MOBILE GENERATION CONTEXT
fprintf('\n------ MOBILE GENERATION CONTEXT ------\n');
fprintf('1G : FDMA - Analog Cellular Systems\n');
fprintf('2G : TDMA/CDMA - Digital Cellular Systems\n');
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fprintf('3G : WCDMA/CDMA - Broadband Mobile Systems\n');
fprintf('4G : OFDMA/SC-FDMA - LTE Systems\n');
fprintf('5G : OFDM-based Multiple Access - 5G NR\n');
%% 8. FINAL COMPARISON TABLE
Results = table(techniques',capacity',efficiency', ...
    'VariableNames',{'Technique','Maximum_Users', ...
    'Spectral_Efficiency_bps_Hz'});
disp(' ');
disp('------ FINAL COMPARISON TABLE ------');
disp(Results);
fprintf('\nSimulation Completed Successfully!\n');
```

output :



